

Full-Validation Detector Adaptation Results

Original unadapted detectors vs 3-epoch adapted detectors

Primary result

**Adaptation recovers a substantial part of the PU-GCN loss,
but no tested adapted PU-GCN arm reaches its reference baseline.**

KITTI validation: 3,769 frames | Car 3D AP_R40 | PointRCNN + CenterPoint

Evidence freeze: 10 September 2026 | Deck: 13 September 2026

Terminology first: PU-GCN was not retrained on KITTI

What 'retrained' means in this deck

- PU-GCN remains frozen at the released PU1K model-100 checkpoint.
- PointRCNN and CenterPoint are adapted from their official pretrained checkpoints.
- The comparison is therefore: original official detector weights vs detector weights adapted for 3 epochs on the matching PU-GCN input.
- Line B also includes schedule-matched adapted baseline detectors.

Direct answer

Did adaptation help?

Yes — all 6 paired PU-GCN arms improved.

Did the adapted models beat baseline?

No — the remaining gaps are 4.58–21.72 AP.

Did 3 epochs converge?

Training loss decreased; validation convergence was not demonstrated.

Completed scope and acceptance criteria

3,712

Adaptation frames

Full KITTI train split for every trained arm

3,769

Validation frames

Complete Line A and Line B generation

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Evaluation arms

Every arm: PASS and frame_count=3,769

AP_R40

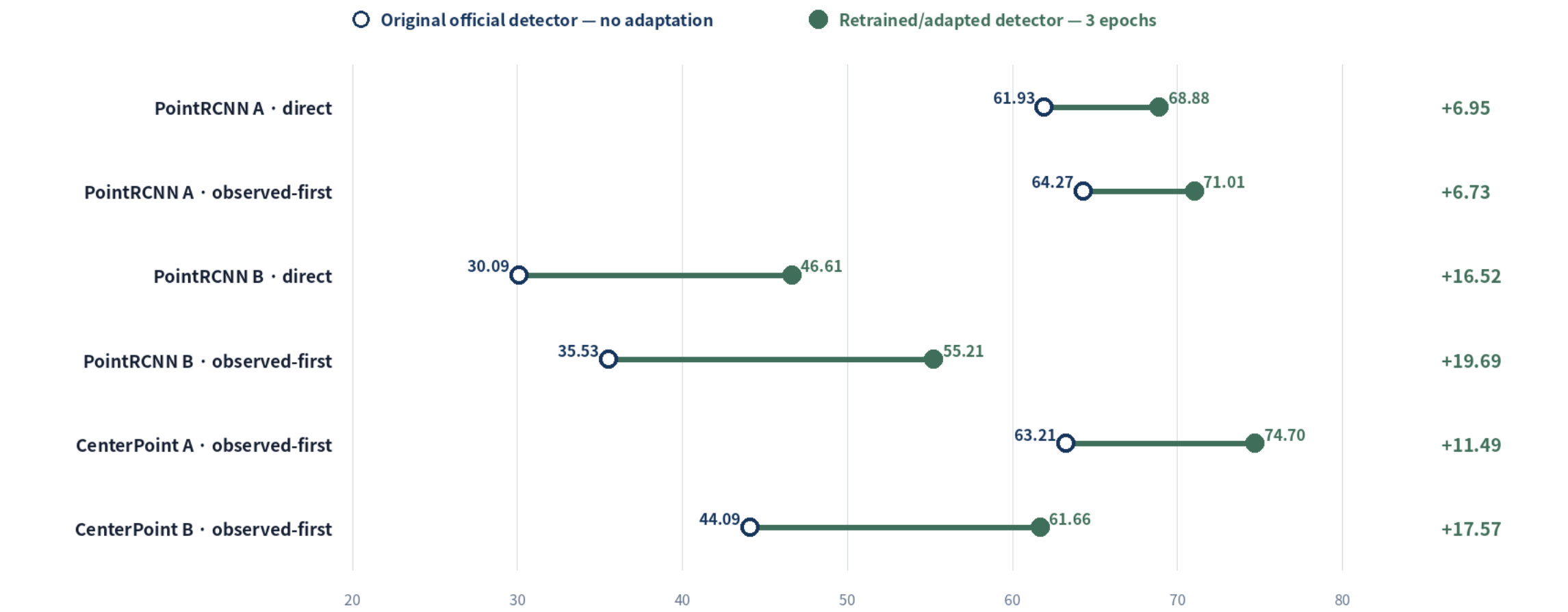
Common metric

Car 3D/BEV; CenterPoint also has 3 classes

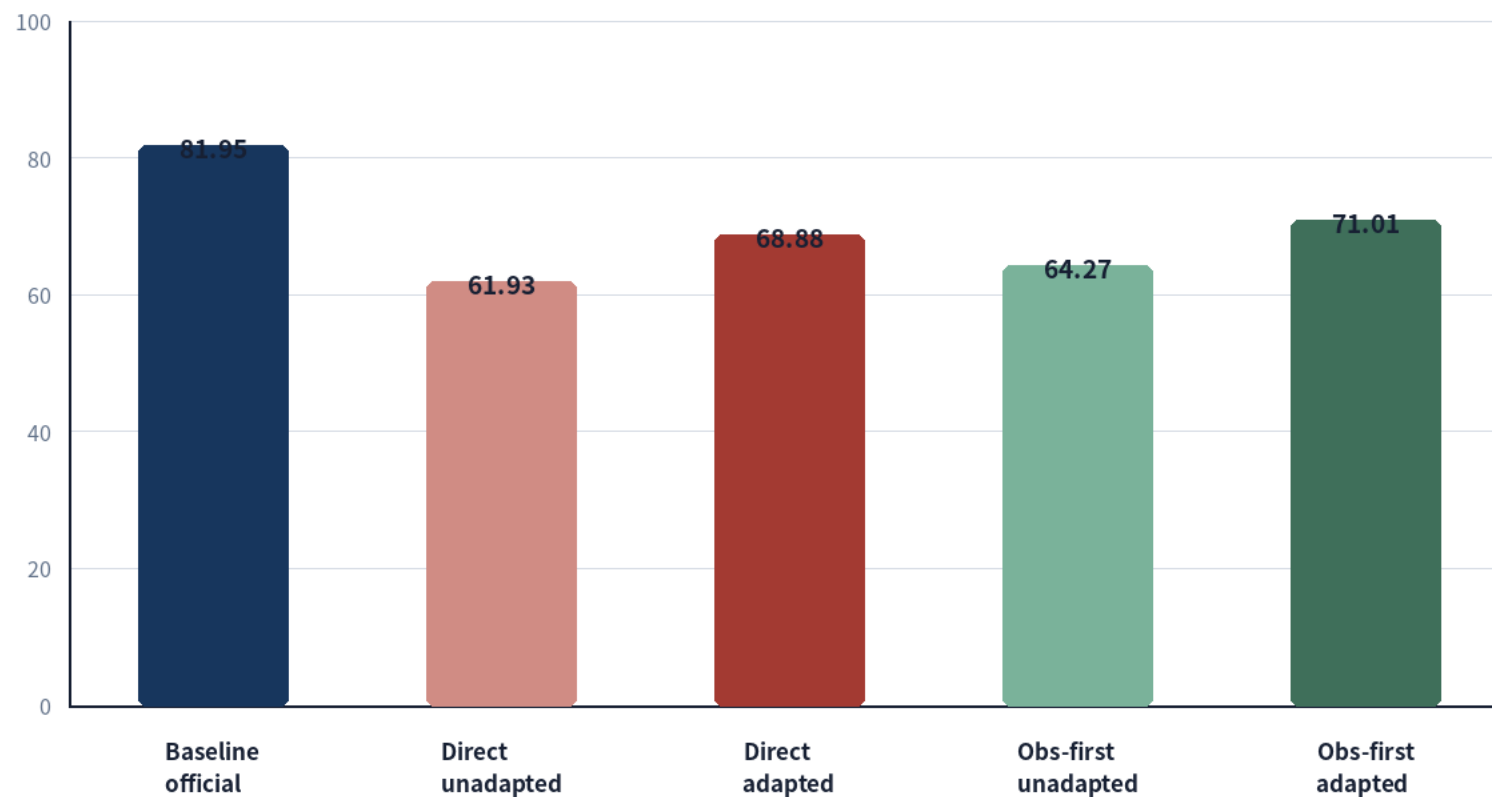
Matrix coverage

Detector	Arms	Official / unadapted	Adapted	Classes
PointRCNN	11	6	5	Car
CenterPoint	9	6	3	Car / Pedestrian / Cyclist
Total	20	12	8	All completed outputs

Direct comparison: original unadapted vs retrained/adapted



PointRCNN · Line A: adaptation narrows the loss



Adaptation gain

Direct: +6.95 AP

Observed-first: +6.73 AP

Reference-baseline gap

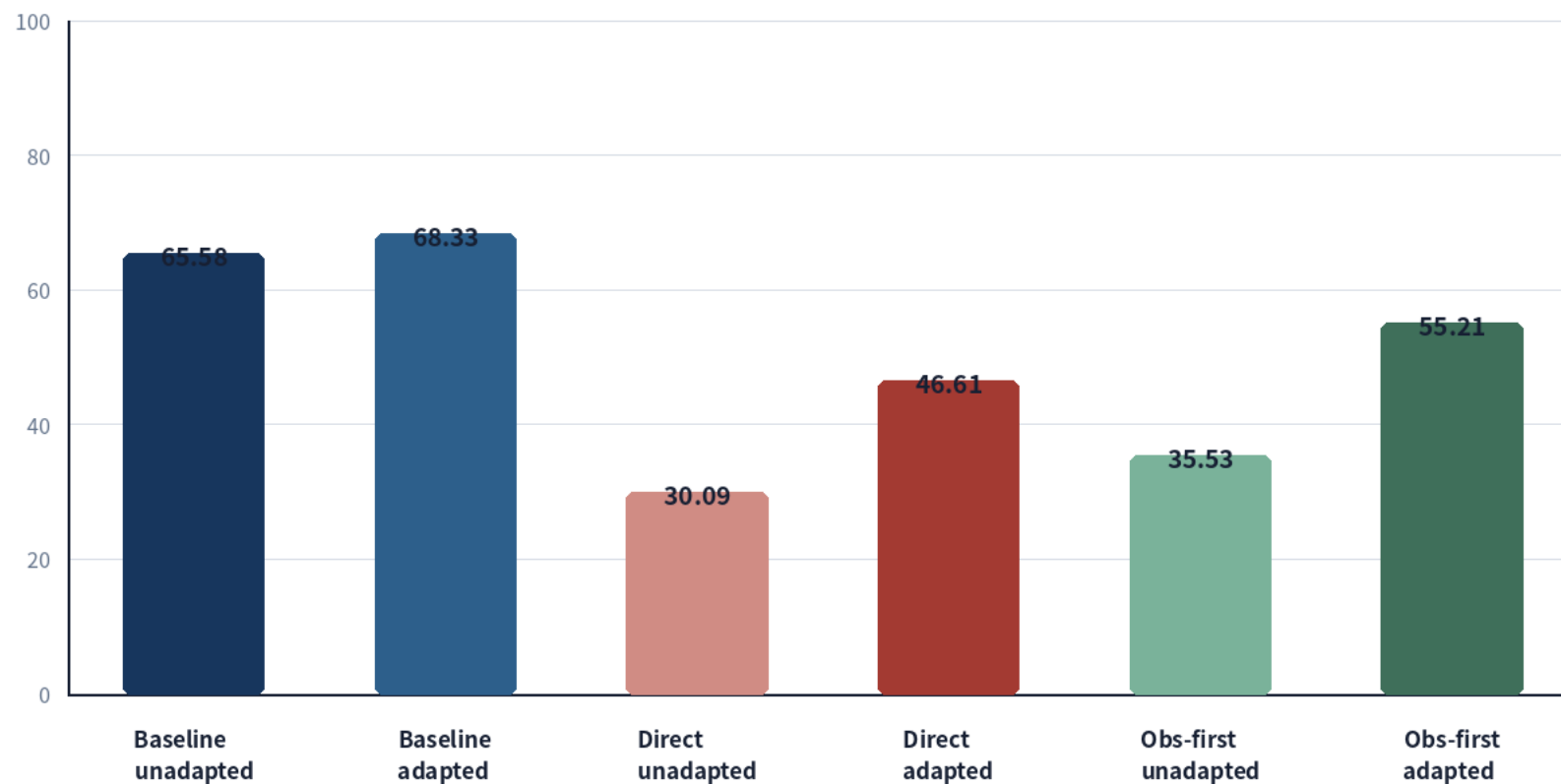
Official baseline: 81.95

Best PU-GCN: 71.01

Gap: -10.95 AP

No schedule-matched adapted original-N baseline exists for Line A; this gap changes both input and weights.

PointRCNN · Line B: the largest recovery is observed-first



Adaptation gain

Direct: +16.52

Observed-first: +19.69

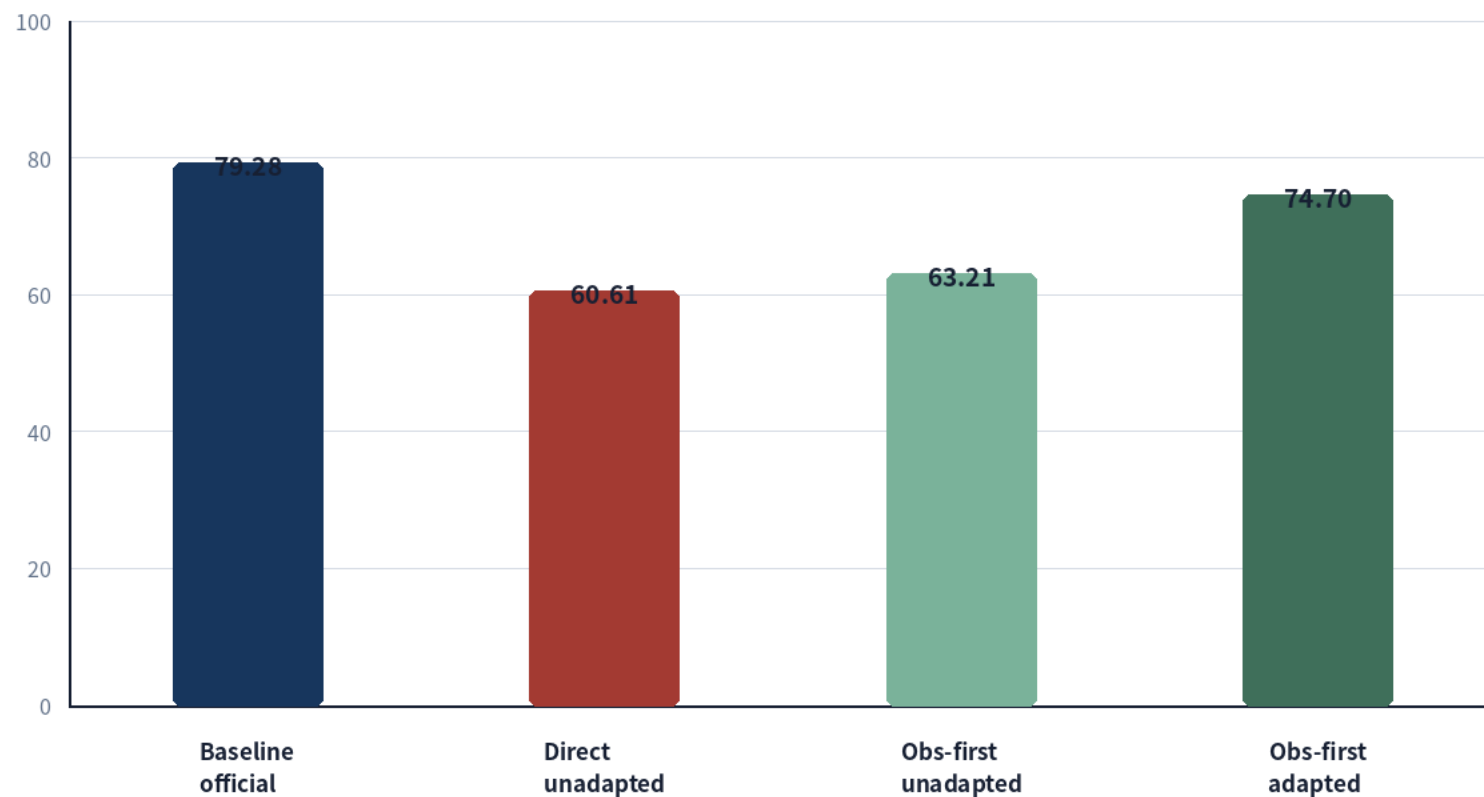
Schedule-matched baseline

Adapted baseline: 68.33

Best PU-GCN: 55.21

Gap: -13.12 AP

CenterPoint · Line A: observed-first approaches baseline



Observed-first adaptation

63.21 → 74.70

Gain: +11.49 AP

Remaining gap

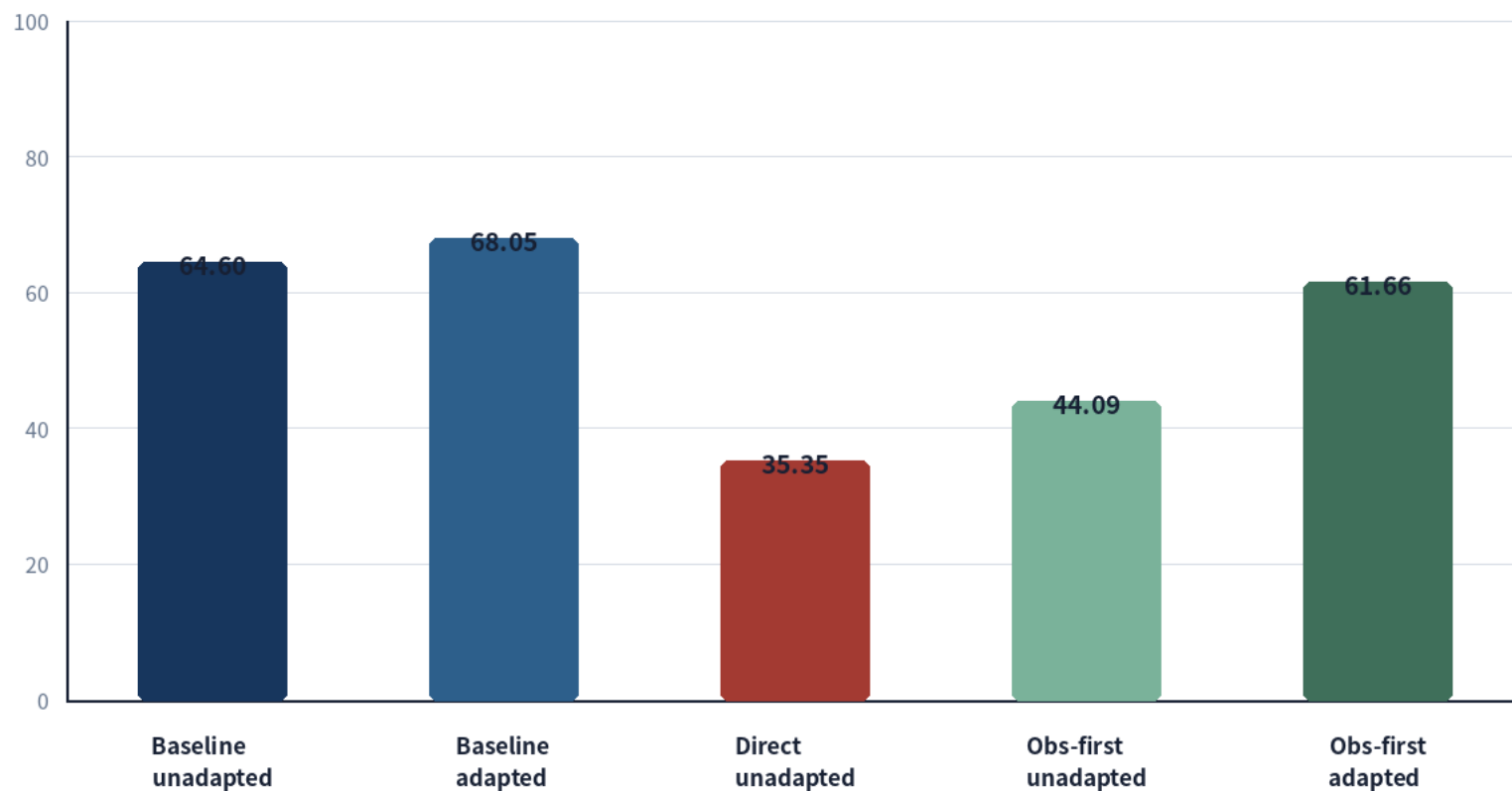
Official baseline: 79.28

Adapted: 74.70

Gap: -4.58 AP

CenterPoint direct-4N adapted was not run and is not imputed.

CenterPoint · Line B: adaptation recovers most of the loss



Observed-first adaptation

44.09 → 61.66

Gain: +17.57 AP

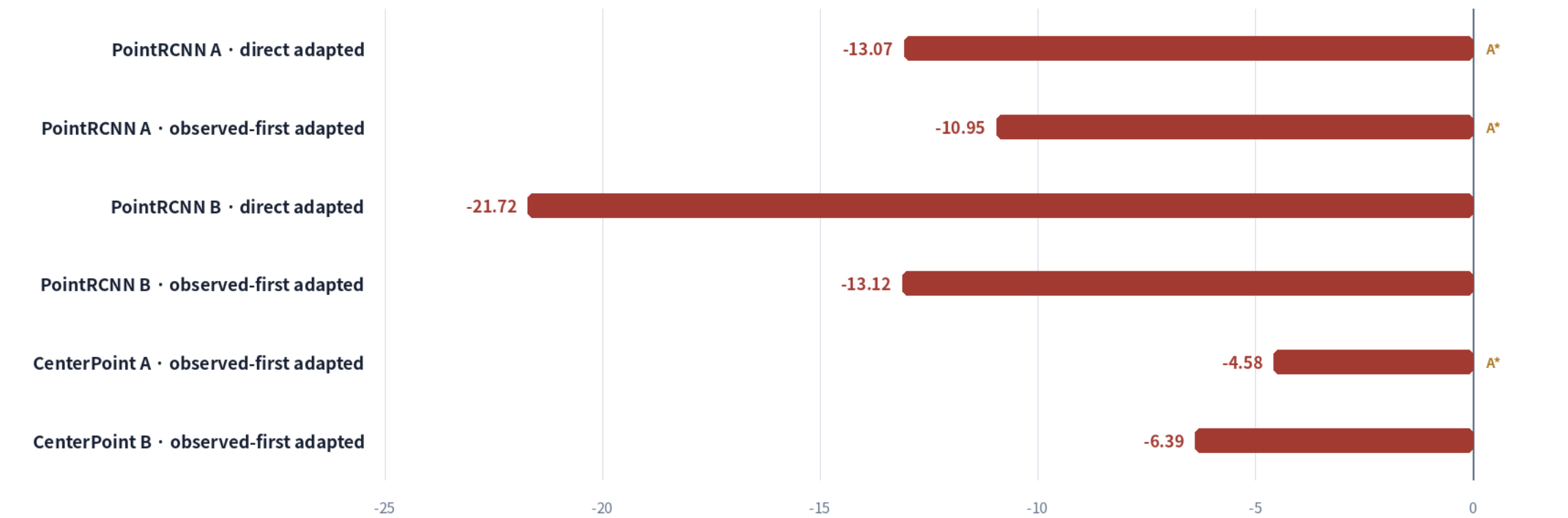
Schedule-matched baseline

Adapted baseline: 68.05

Adapted PU-GCN: 61.66

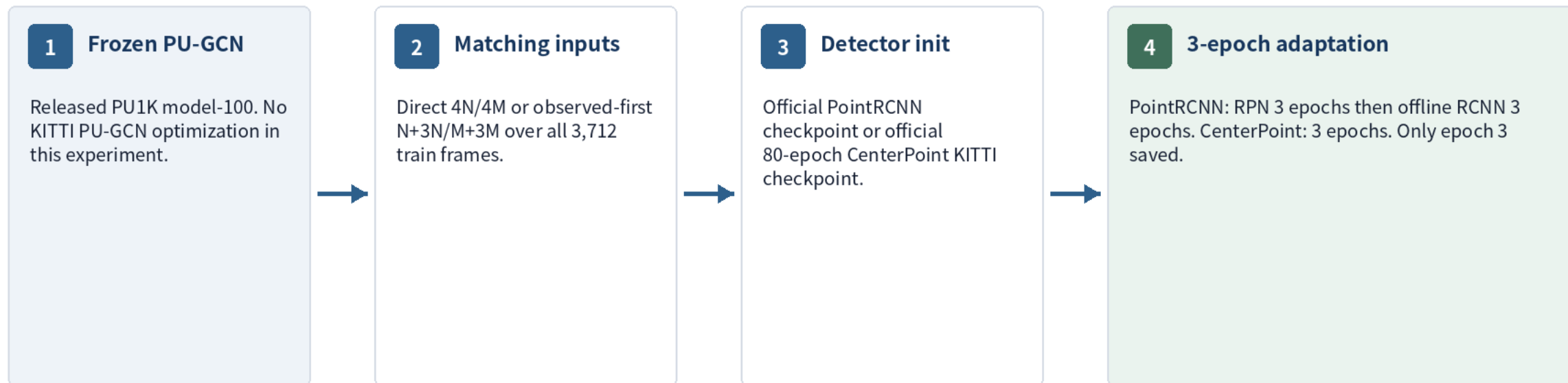
Gap: -6.39 AP

Adaptation helps, but 4.58–21.72 AP remains



A*: no schedule-matched adapted original-N baseline was trained; the displayed Line A delta uses the official baseline and changes both input and weights.

What was frozen and what was adapted



Interpretation boundary

These are short detector-adaptation runs from pretrained checkpoints, not from-scratch training and not evidence that PU-GCN itself was retrained.

Why exactly 3 epochs? The recorded evidence and its limit

Recorded facts

- Both training wrappers set the default epoch count to 3.
- Every executed adaptation protocol records epochs=3.
- The same short schedule was applied across arms for controlled comparison.
- Only epoch 3 was checkpointed and evaluated.

Operational interpretation

- The detectors start from strong pretrained checkpoints, so the run is adaptation rather than full retraining.
- Three epochs function as a fixed, compute-bounded adaptation budget.
- This is a reasonable interpretation of the implementation, not a documented empirical selection study.

What cannot be claimed

- No experiment record explains why 3 was chosen instead of 6 or 10.
- No validation curve selected epoch 3.
- No early-stopping rule or plateau criterion was used.
- Therefore 3 epochs is a pre-fixed schedule, not a demonstrated optimum.

Code evidence: 3 epochs and epoch-3-only checkpointing

```
scripts/run_pointcnn_full_train_stage.py
```

```
# PointRCNN wrapper
```

```
parser.add_argument("--epochs",
```

```
                    type=int, default=3)
```

```
...
```

```
"--epochs", str(args.epochs),
```

```
"--ckpt_save_interval",
```

```
str(args.epochs),
```

```
scripts/run_centerpoint_full_train.py
```

```
# CenterPoint wrapper
```

```
parser.add_argument("--epochs",
```

```
                    type=int, default=3)
```

```
...
```

```
"--epochs", str(args.epochs),
```

```
"--ckpt_save_interval",
```

```
str(args.epochs),
```

```
"--num_epochs_to_eval", "0",
```

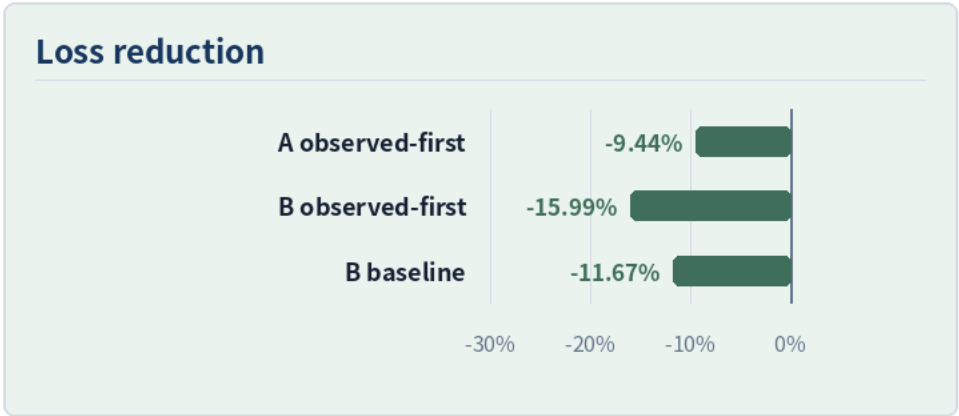
Observed artifacts: 10 PointRCNN epoch-3 checkpoints + 3 CenterPoint epoch-3 checkpoints; no epoch-1/2 checkpoints exist in these runs.

PointRCNN: complete 3-epoch optimization audit

Arm	Stage	Epoch 1	Epoch 2	Epoch 3	E1→E3	Outliers
A direct	RPN	1.6241	1.4449	1.3475	-17.03%	0
A direct	RCNN	1.0430	0.9963	0.9596	-8.00%	7
A obs-first	RPN	1.5491	1.4196	1.2941	-16.46%	0
A obs-first	RCNN	1.0128	1.0010	0.9680	-4.42%	12
B baseline	RPN	1.6964	1.4334	1.2901	-23.95%	2
B baseline	RCNN	0.9556	0.9451	0.9159	-4.16%	13
B direct	RPN	2.2404	1.8157	1.6218	-27.61%	1
B direct	RCNN	1.3049	1.2420	1.2051	-7.65%	12
B obs-first	RPN	2.0874	1.7537	1.5694	-24.82%	0
B obs-first	RCNN	1.2545	1.2048	1.1684	-6.86%	0

CenterPoint: complete 3-epoch optimization audit

Arm	E1	E2	E3	E1→E3	Final LR
A observed-first	2.33	2.21	2.11	-9.44%	3.066e-09
B observed-first	3.69	3.29	3.10	-15.99%	3.066e-09
B baseline	3.00	2.81	2.65	-11.67%	3.066e-09



Result

All three mean-loss series decrease from epoch 1 to epoch 3.

This proves optimization progress; it does not prove that validation AP has plateaued or that epoch 3 is optimal.

Convergence verdict: all losses fell; validation convergence is unproven

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Training-loss series decreased

10 PointRCNN stages + 3 CenterPoint arms

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Only epoch-3 checkpoints

No epoch-1 or epoch-2 checkpoints retained

0

Per-epoch validation AP points

No AP curve exists for epoch selection

NO

Convergence demonstrated?

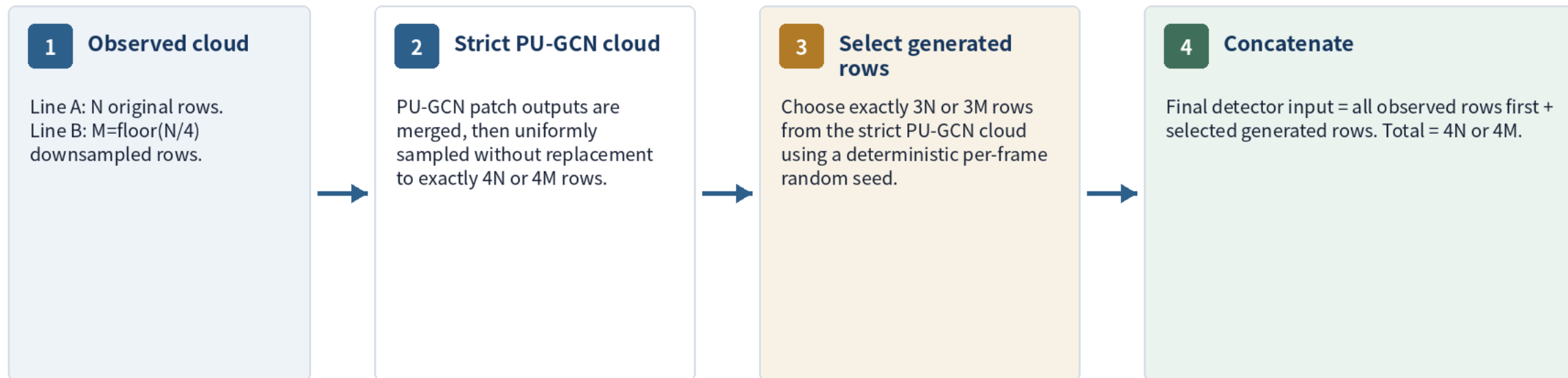
Schedule completed; optimum/plateau not established

The only defensible statement

The three-epoch adaptation schedule completed successfully and every recorded training-loss series decreased.

Because only epoch 3 was saved and there is no per-epoch validation AP, the current records cannot establish convergence or select epoch 3 as the optimum.

What '3N' means: observed-first composition



Important

3N is not 'the three new children per input anchor.' It is a second frame-level sample of $3N$ rows from the already constructed strict $4N$ PU-GCN output.

Exact 3N/3M selection code

Deterministic uniform sampling without replacement

BASE_SEED = 20260718

```
def stable_seed(*parts):
    digest = hashlib.sha256("|".join(str(p) for p in parts)
                                .encode("utf-8")).digest()
    return int.from_bytes(digest[:8], "little") & 0xFFFFFFFF

expected = 4 * observed.shape[0]
assert predicted.shape[0] == expected
seed = stable_seed(BASE_SEED, "e1", reference_token, frame)
rng = np.random.default_rng(seed)
selected = rng.choice(expected, size=3 * observed.shape[0],
                      replace=False)
final = np.concatenate((observed, predicted[selected]), axis=0)
```

Line A reference_token: original_x4_pu_gcn | Line B: downsampled_x4_pu_gcn

The SHA-256-derived seed makes selection repeatable per frame and distinct across lines/reference tokens.

Selection rule: what it uses and what it does not use

The actual rule

- Candidate set: every row in the strict 4N/4M PU-GCN output.
- Sample size: exactly 3N/3M.
- Distribution: uniform over row indices.
- Replacement: false.
- Determinism: stable hash of base seed, token, and frame ID.
- Output order: all observed rows, then selected generated rows.

Not used by this rule

- No confidence score or uncertainty.
- No distance-to-observed ranking.
- No FPS, curvature, voxel quota, or foreground weighting.
- No coordinate deduplication.
- No guarantee that selected generated coordinates are geometrically distinct from observed coordinates.
- 'Generated' is a provenance label, not a set-difference operation.

3N/3M audit evidence: every validation frame passes

3,769 / 3,769

Line A rows audited

PASS; predicted=4N, selected=3N, output=4N

3,769 / 3,769

Line B rows audited

PASS; predicted=4M, selected=3M, output=4M

TRUE

Observed prefix exact

All frames, both lines

TRUE

Generated suffix exact

All frames, both lines

Example	Observed	Predicted	Selected	Output	Seed	Status
A • frame 000001	120,268	481,072	360,804	481,072	1091044475	PASS
B • frame 000001	30,067	120,268	90,201	120,268	2918483960	PASS

Each manifest row also stores selected_indices_sha256 and output_float32_sha256, enabling byte-level reproduction checks.

From a PU-GCN patch to strict frame-level 4×

Network-level principle

- Each 2,048-point patch is centered and normalized to a unit sphere.
- The graph network extracts features and NodeShuffle upsamples them.
- A coordinate head predicts 3D residuals.
- Each input anchor is tiled four times and the residuals are added, producing 8,192 rows per patch.
- Predictions are transformed back to metric coordinates.

PU-GCN residual output

```
outputs = tf.squeeze(coord, [2])
outputs += tf.reshape(tf.tile(
    tf.expand_dims(inputs, 2),
    [1, 1, self.up_ratio, 1]), ...)
```

Strict frame-level 4×

```
target_points = input_count * 4
idx = rng.choice(raw_count,
                 size=target_points,
                 replace=False)
strict_xyz = raw_xyz[idx]
intensity = nearest_observed_intensity(strict_xyz)
```

Complete PointRCNN Car results

Line	Input / weights	3D E	3D M	3D H	BEV M	Status
A	baseline N / unadapted	92.43	81.95	77.85	88.86	PASS
A	direct 4N / unadapted	84.74	61.93	57.79	71.45	PASS
A	direct 4N / adapted	85.17	68.88	64.67	80.52	PASS
A	observed N + predicted 3N / unadapted	85.04	64.27	60.09	73.79	PASS
A	observed N + predicted 3N / adapted	86.71	71.01	66.79	81.03	PASS
B	baseline M / unadapted	85.20	65.58	61.28	74.92	PASS
B	baseline M / adapted	84.39	68.33	64.26	77.97	PASS
B	direct 4M / unadapted	46.47	30.09	25.80	38.74	PASS
B	direct 4M / adapted	68.26	46.61	40.25	58.20	PASS
B	observed M + predicted 3M / unadapted	55.02	35.53	30.94	43.86	PASS
B	observed M + predicted 3M / adapted	74.20	55.21	49.06	67.30	PASS

Complete CenterPoint Car results

Line	Input / weights	3D E	3D M	3D H	BEV M	Status
A	baseline N / unadapted	88.39	79.28	76.74	88.43	PASS
A	direct 4N / unadapted	81.89	60.61	57.91	70.68	PASS
A	observed N + predicted 3N / unadapted	83.31	63.21	61.04	72.83	PASS
A	observed N + predicted 3N / adapted	86.82	74.70	72.66	84.93	PASS
B	baseline M / unadapted	81.40	64.60	59.97	77.31	PASS
B	baseline M / adapted	83.75	68.05	64.63	80.20	PASS
B	direct 4M / unadapted	56.42	35.35	31.20	47.53	PASS
B	observed M + predicted 3M / unadapted	66.92	44.09	40.03	55.54	PASS
B	observed M + predicted 3M / adapted	80.41	61.66	57.53	73.67	PASS

CenterPoint direct-adapted was not executed; it is intentionally absent from results and charts.

CenterPoint: complete Moderate 3D AP_R40 for all classes

Line	Input / weights	Car	Pedestrian	Cyclist
A	baseline N / unadapted	79.28	50.65	64.61
A	direct 4N / unadapted	60.61	45.08	42.83
A	observed N + predicted 3N / unadapted	63.21	45.98	49.45
A	observed N + predicted 3N / adapted	74.70	48.42	57.70
B	baseline M / unadapted	64.60	24.57	15.46
B	baseline M / adapted	68.05	47.44	42.07
B	direct 4M / unadapted	35.35	24.89	15.28
B	observed M + predicted 3M / unadapted	44.09	29.03	20.02
B	observed M + predicted 3M / adapted	61.66	44.06	33.56

The class-level conclusion is restricted to CenterPoint and is not extrapolated to PointRCNN.

Code provenance: exact files and SHA-256 evidence

Evidence item	Path	SHA-256
3N selector	scripts/prepare_centerpoint_observed_first_train.py	f5e521219f8d49b8fed8d9d40192749378b02a0f54888b92e949d041db1f4f42
PointrCNN trainer	scripts/run_pointrcnn_full_train_stage.py	4b177d0dd27769a5f562ec45060e09537654965ab94936d9bd2df8f43b43828a
CenterPoint trainer	scripts/run_centerpoint_full_train.py	a71cbac6817271e53c19828cefbcdabc5ae1ad9d9c9bf5d4e665d16f211801ee6
Convergence audit	reports/convergence_audit.json	3326d80c8916023936b9be7ad2b27f8469a8c9f32079fe8107ed60e8fadeacca
Full result matrix	reports/full_val_detector_matrix.csv	68606b1062b93c8ae7ac1965f62e53221df2851eb533883e69ecb3632f3c63bd

Snapshot package

[reports/pugcn_full_val_sources.tar.gz](#) [reports/source_manifest.json](#) [reports/experiment_source_full.md](#)

The package records byte-exact local code used by the experiment; it does not imply that every file equals the upstream author repository.

Evidence index, reproduction entry point, and final verdict

Reproduction entry point

```
cd /home/ra87racy/projects/baseline_detectors/PointRCNN
bash scripts/run_pugcn_detector_adaptation_full_val_20260908.sh all
```

Primary evidence

- reports/full_val_detector_matrix.csv
- reports/all_classes_ap_r40.csv
- reports/convergence_audit.json
- inputs/line_[ab]_pugcn_observed_first_manifest.csv
- reports/protocol_and_code_zh.md

Final verdict

Adaptation effect

Positive in all 6 paired PU-GCN arms

Baseline recovery

Incomplete: every tested adapted PU-GCN arm remains below baseline

Convergence

Optimization progressed; validation convergence was not demonstrated